

Cenxi Zhang

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EDUCATION

Qingdao University of Technology

Sep 2023 - Jun 2027

Bachelor of Engineering in Building Electrical and Intelligent

GPA: 4.52 / 5.0 (Ranking: 3/70)

RESEARCH INTERESTS

LLM Agents; Multi-Agent Systems; AI for Embodied Interaction; Agentic Simulation; Intelligent Robotic Systems

RESEARCH, TEACHING, AND TECHNICAL EXPERIENCE

Innovation Laboratory, School of Information and Control Engineering

Jan 2024 - Present

Student Technical Lead | Qingdao University of Technology

- Served as student technical lead for the Innovation Laboratory, coordinating AI-oriented technical training, project mentoring, and competition preparation across LLM agents, AI fundamentals, computer vision, embedded systems, and control systems.
- Mentored undergraduate students in building AI-driven perception-decision-control systems from first principles, covering model-based perception, OpenCV-based vision, PID tuning, Kalman filtering, STM32 development, and real-time debugging.
- Developed hands-on teaching workflows for AI and embedded system projects, helping students connect algorithm design, system integration, debugging, and competition-oriented engineering practice.

Friction-Aware Cooperative ADRC-NN Control for Multi-DOF Manipulators

Sep 2025 - Present

Core Developer | MATLAB / Simulink / Python / Control Theory / Neural Networks

- Designed a cooperative ADRC and neural-network strategy for compensating nonlinear friction disturbances in multi-DOF manipulators.
- Implemented the ESO-NN control architecture and completed data collection, cleaning, feature engineering, and simulation-based validation.
- Compared the method with PID and pure ADRC baselines, focusing on tracking accuracy, robustness, and low-speed friction compensation.

32-hour AI Training Course

Apr 2026 - Jul 2026

Developer and Presenter | Neural Networks / CNNs / Transformers / LLM Agents / Generative Models

- Developed and presented a 32-hour AI training course for 93 master's students, covering machine learning basics, neural networks, CNNs, NLP, Transformers, LLM agents, generative models, and applied AI systems.
- Created project-oriented notebooks, datasets, and code examples connecting mathematical intuition, model architecture, implementation, and deployment.
- Published course materials on GitHub and Bilibili, including lecture notebooks, code walkthroughs, and project demonstrations: github.com/zcxixixi/ai-course.

HKU Data Intelligence Laboratory nanobot Open-Source Project

Feb 2026 - Apr 2026

Open-Source Contributor | Python / CLI / IMAP / SMTP / GitHub

- Contributed multiple pull requests to the HKU Data Intelligence Laboratory nanobot open-source project.
- Extended external communication capabilities for agents and improved CLI reliability for CJK text workflows.
- Refined interaction-stack logic for command handling and user-agent communication flow.

Embedded Vision Measurement and Control System

Aug 2025

Team Leader, National Undergraduate Electronics Design Contest | C / Python / YOLO / OpenCV / Serial Communication | Demo: youtu.be/vVcSWguqrLc

- Led a three-member team to build a full-stack embedded vision measurement and control system from scratch within a four-day, three-night closed national contest, winning National First Prize, ranking 3rd in Shandong Province, and placing among the national top 1%.
- Developed the end-to-end vision pipeline, integrating YOLO-based object recognition, OpenCV geometric measurement, camera calibration, and perspective transformation for real-time target localization.
- Designed a side-length-variation-based differential algorithm to separate overlapping targets and improve coordinate mapping reliability.
- Implemented serial communication between the vision module and the control unit, enabling real-time data transfer for downstream control and actuation.

LLM-Based Intelligent Wheel-Legged Interactive Robot

Jun 2025

Core Developer, Embedded Chip and System Design Competition | LVGL / OpenCV / YOLOv8 / STM32 / IoT / LLM | Demo: youtu.be/dhbkG9M1-zg

- Developed an LLM-based intelligent wheel-legged interactive robot for the Embedded Chip and System Design Competition, winning First Prize in the East China Division.
- Integrated LLM-driven interaction logic with gesture recognition, light perception, and IoT control to support multimodal human-robot interaction.
- Built STM32-based control modules for wheel-legged mobility, sensor integration, and real-time hardware interaction.
- Designed an LVGL-based interface for human-robot interaction, system status display, and interactive control.

Selected Embedded and Robotic Systems

Sep 2024 - Jun 2025

Core Developer | STM32 / Control Systems / Computer Vision / Robotic Actuation

- Wind Pendulum Control System: developed a dual-axis control system using PID and feedforward compensation for disturbance-resistant oscillation control. Demo: youtu.be/A5QPHtyPR2g.
- High-Precision Line-Following Car: built an embedded PID control system for stable line detection, path tracking, and adaptive speed regulation under competition-oriented conditions. Demo: youtu.be/UzqoWBQiwA0.
- Two-Wheel Self-Balancing Car: implemented Kalman-filter-based attitude estimation, cascaded PID control, and motor regulation for autonomous balancing and motion control.
- Autonomous Tic-Tac-Toe Robotic Device: integrated board-state recognition, minimax-based decision logic, and robotic arm actuation to complete human-machine gameplay.

HONORS AND AWARDS

First-Class Scholarship for Outstanding Students, Qingdao University of Technology, 2024, 2025, and 2026.

National First Prize, National Undergraduate Electronics Design Contest, University Group, 2025.

First Prize, Embedded Chip and System Design Competition, East China Division, 2025.

Third Prize, Lanqiao Cup National Finals, Embedded Design and Development, 2025.

Discipline Competition Scholarship, Qingdao University of Technology, 2025.

First Prize, National Undergraduate Electronics Design Contest, Shandong Division, 2024.

SKILLS

AI Engineering: LLM agent architecture, multi-agent workflow design, tool-augmented reasoning, RAG concepts, LangGraph, prompt and workflow engineering, Claude Code, Codex.

Programming and Systems: Python, C, MATLAB, TypeScript, FastAPI, Next.js, Git/GitHub, GitHub Actions, Linux, serial communication.

Embedded and Control: STM32, embedded C, PID and cascaded control, Kalman filtering, ADRC, MATLAB/Simulink, sensor integration, real-time debugging, on-device deployment.

Computer Vision and Robotics: YOLO, OpenCV, camera calibration, perspective transformation, geometric measurement, perception-decision-control integration, robotic actuation.

Languages: Chinese (native), English (working proficiency).



青岛理工大学学生中文成绩单

学号: 202301050482

姓名: 张涔熙

性别: 男

出生日期: 20050302

班级: 建电231

院部: 信息与控制工程学院

专业: 建筑电气与智能化

学制: 4

课程名称	学分	成绩	绩点	补考重修	学年	学期	课程名称	学分	成绩	绩点	补考重修	学年	学期
大学计算机 (A)	2.0	90	4.5		23-24	1	数字电子技术基础	3.0	98	5.0		24-25	1
大学生心理健康	2.0	88	4.3		23-24	1	物理实验下	0.5	80	3.5		24-25	1
大学体育 I (篮球)	1.0	91	4.6		23-24	1	形势与政策3	0.25	90	4.5		24-25	1
高等数学A(上)	5.0	94	4.9		23-24	1	学术英语 I	2.0	91	4.6		24-25	1
工程图学基础B	3.0	86	4.1		23-24	1	职业生涯规划——体验式学习	1.0	99	5.0		24-25	1
公益类劳动实践1	0	99	5.0		23-24	1	创新创业基础	2.0	89	4.4		24-25	2
建筑电气与智能化专业导论	1.0	良好	3.5		23-24	1	大话机器人	1.0	95	5.0		24-25	2
军事理论课	2.0	96	5.0		23-24	1	大学体育IV (篮球)	1.0	93	4.8		24-25	2
军事训练	2.0	95	5.0		23-24	1	电力电子技术	2.5	95	5.0		24-25	2
劳动教育基础1	0	99	5.0		23-24	1	工程电磁场	2.0	84	3.9		24-25	2
习近平新时代中国特色社会主义思想概论	3.0	82	3.7		23-24	1	计算机工程实习	1.0	90	4.5		24-25	2
形势与政策1	0.25	96	5.0		23-24	1	建筑环境学	2.0	89	4.4		24-25	2
中国近现代史纲要	3.0	88	4.3		23-24	1	建筑通信与网络	2.0	87	4.2		24-25	2
自动化工程通用英语 I	4.0	89	4.4		23-24	1	马克思主义基本原理	3.0	92	4.7		24-25	2
程序设计基础A (C语言)	3.0	94	4.9		23-24	2	微型计算机系统*	2.5	96	5.0		24-25	2
程序实习 (C)	2.0	优秀	4.5		23-24	2	微型计算机系统课程设计	1.0	优秀	4.5		24-25	2
大学体育 II (篮球)	1.0	94	4.9		23-24	2	形势与政策4	0.25	98	5.0		24-25	2
大学物理A(上)	3.0	94	4.9		23-24	2	学术英语 II	2.0	87	4.2		24-25	2
德语影视欣赏	2.0	87	4.2		23-24	2	中华文明史	1.5	97	5.0		24-25	2
电路原理	4.0	88	4.3		23-24	2	自动控制理论	3.5	95	5.0		24-25	2
电路原理实验课	0.5	87	4.2		23-24	2	BIM应用基础	2.0	97	5.0		25-26	1
高等数学A(下)	6.0	89	4.4		23-24	2	PLC综合实训	2.0	89	4.4		25-26	1
公益类劳动实践	0.2	95	5.0		23-24	2	传感器与检测技术	3.0	91	4.6		25-26	1
思想道德与法治	3.0	95	5.0		23-24	2	电机与拖动	3.0	81	3.6		25-26	1
物理实验上	1.0	73	2.8		23-24	2	电气控制与PLC应用	2.5	83	3.8		25-26	1
线性代数	2.5	95	5.0		23-24	2	电气照明技术	2.0	93	4.8		25-26	1
形势与政策2	0.25	86	4.1		23-24	2	电气照明技术课程设计	1.0	优秀	4.5		25-26	1
职业生涯规划	0.5	93	4.8		23-24	2	电子实习	1.0	优秀	4.5		25-26	1
中国共产党隐蔽战线的斗争	1.5	95	5.0		23-24	2	建筑供电系统	3.0	94	4.9		25-26	1
自动化工程通用英语 II	4.0	91	4.6		23-24	2	建筑供电系统课程设计	1.0	优秀	4.5		25-26	1
大学体育 III (篮球)	1.0	97	5.0		24-25	1	建筑设备	2.0	95	5.0		25-26	1
大学物理A(下)	3.0	94	4.9		24-25	1	形势与政策5	0.25	93	4.8		25-26	1
电子技术实验课	1.0	89	4.4		24-25	1	中外经典影视音乐欣赏	2.0	93	4.8		25-26	1
复变函数与积分变换	2.0	95	5.0		24-25	1	专业实践类劳动实践	0.2	76	3.1		25-26	1
概率论与数理统计	2.5	86	4.1		24-25	1							
金工实习	1.0	良好	3.5		24-25	1							
毛泽东思想和中国特色社会主义理论体系概论	3.0	90	4.5		24-25	1							
模拟电子技术基础	3.0	91	4.6		24-25	1							
认识实习	2.0	90	4.5		24-25	1							





Student Transcript of Qingdao University of Technology

Student No.: 202301050482

Name: Zhang Cenxi

Gender: Male

Date of Birth: 20050302

Class: Building Electrical and Intelligent 231

School: School of Information and Control Engineering

Major: Building Electrical and Intelligent

Length of Schooling: 4 Years

Course Name	Credit	Score	Grade Point	Resit Retake	Year	Term	Course Name	Credit	Score	Grade Point	Resit Retake	Year	Term
University Computer Science (A)	2.0	90	4.5		23-24	1	Fundamentals Digital Electronic Technique	3.0	98	5.0		24-25	1
Mental Health of College Students	2.0	88	4.3		23-24	1	Physical Experiments (Part 2)	0.5	80	3.5		24-25	1
University PE I (Basketball)	1.0	91	4.6		23-24	1	Situation and Policy 3	0.25	90	4.5		24-25	1
Advanced Mathematics A (Part 1)	5.0	94	4.9		23-24	1	Academic English I	2.0	91	4.6		24-25	1
Fundamentals of Engineering Graphics B	3.0	86	4.1		23-24	1	Career Planning - Experiential Learning	1.0	99	5.0		24-25	1
Public Welfare Labor Practice I	0	99	5.0		23-24	1	Fundamentals of Innovation and Entrepreneurship	2.0	89	4.4		24-25	2
Introduction to Building Electrical and Intelligent Engineering	1.0	Good	3.5		23-24	1	Big Talk Robot	1.0	95	5.0		24-25	2
Military Theory	2.0	96	5.0		23-24	1	University PE IV (Basketball)	1.0	93	4.8		24-25	2
Military Training	2.0	95	5.0		23-24	1	Power Electronics	2.5	95	5.0		24-25	2
Fundamentals of Labor Education	0	99	5.0		23-24	1	Engineering Electromagnetic Field	2.0	84	3.9		24-25	2
An Introduction to Xi Jinping's Thought on Socialism with Chinese Characteristics for a New Era	3.0	82	3.7		23-24	1	Computer Engineering Practice	1.0	90	4.5		24-25	2
Situation and Policy 1	0.25	96	5.0		23-24	1	Building Environment	2.0	89	4.4		24-25	2
Outline of Chinese Modern History	3.0	88	4.3		23-24	1	Building Communications and Networks	2.0	87	4.2		24-25	2
General English for Automation Engineering I	4.0	89	4.4		23-24	1	Basic Principles of Marxism	3.0	92	4.7		24-25	2
Fundamentals of Programming A (C Language)	3.0	94	4.9		23-24	2	Microcomputer System*	2.5	96	5.0		24-25	2
Program Internship (C)	2.0	Excellent	4.5		23-24	2	Course Design of Microcomputer System	1.0	Excellent	4.5		24-25	2
University PE II (Basketball)	1.0	94	4.9		23-24	2	Situation and Policy 4	0.25	98	5.0		24-25	2
University Physics A (Part 1)	3.0	94	4.9		23-24	2	Academic English II	2.0	87	4.2		24-25	2
Appreciation of German Film and Television	2.0	87	4.2		23-24	2	History of Chinese Civilization	1.5	97	5.0		24-25	2
Circuit Principle	4.0	88	4.3		23-24	2	Automatic Control Theory	3.5	95	5.0		24-25	2
Circuit Principle Laboratory Class	0.5	87	4.2		23-24	2	Fundamentals of BIM Applications	2.0	97	5.0		25-26	1
Advanced Mathematics A (Part 2)	6.0	89	4.4		23-24	2	PLC Comprehensive Practical Training	2.0	89	4.4		25-26	1
Public Welfare Labor Practice	0.2	95	5.0		23-24	2	Sensors and Detection Technology	3.0	91	4.6		25-26	1
Ideological Ethics and the Rule of Law	3.0	95	5.0		23-24	2	Electrical Machinery and Towage	3.0	81	3.6		25-26	1
Physical Experiments (Part 1)	1.0	73	2.8		23-24	2	Electrical Control and PLC Applications	2.5	83	3.8		25-26	1
Linear Algebra	2.5	95	5.0		23-24	2	Electrical Lighting Technology	2.0	93	4.8		25-26	1
Situation and Policy 2	0.3	86	4.1		23-24	2	Course Design of Electrical Lighting Technology	1.0	Excellent	4.5		25-26	1
Career Planning	0.5	93	4.8		23-24	2	Electronic Practice	1.0	Excellent	4.5		25-26	1
The Struggle on the Covert Front of the CPC	1.5	95	5.0		23-24	2	Building Power Supply System	3.0	94	4.9		25-26	1
General English for Automation Engineering II	4.0	91	4.6		23-24	2	Course Design for Building Power Supply Systems	1.0	Excellent	4.5		25-26	1
University PE III (Basketball)	1.0	97	5.0		24-25	1	Building Implements	2.0	95	5.0		25-26	1
University Physics A (Part 2)	3.0	94	4.9		24-25	1	Situation and Policy 5	0.25	93	4.8		25-26	1
Electronic Technology Laboratory Course	1.0	89	4.4		24-25	1	Appreciation of Classic Chinese and Foreign Film and Television Music	2.0	93	4.8		25-26	1
Complex Function and Integral Transform	2.0	95	5.0		24-25	1	Professional and Practical Labor Practice	0.2	76	3.1		25-26	1
Probability Theory and Mathematical Statistics	2.5	86	4.1		24-25	1							
Metalworking Practice	1.0	Good	3.5		24-25	1							
An Introduction to Mao Zedong Thought and the Theoretical System of Socialism with Chinese Characteristics	3.0	90	4.5		24-25	1							
Fundamentals of Analog Electronic Technology	3.0	91	4.6		24-25	1							
Cognition Practice	2.0	90	4.5		24-25	1							



Academic Affairs Office:

2026-05-12



青島理工大學
QINGDAO UNIVERSITY OF TECHNOLOGY

平均学分绩点及排名证明

张潞熙，男，学号：202301050482，系青岛理工大学
信息与控制工程学院建筑电气与智能化专业2023级本科
生。截至当前修完课程的平均学分绩点为4.55，年级专
业排名为3/70。此证明仅限用于该学生留学申请使用。

特此证明



注：

（一）课程绩点的换算方法

百分制	成绩	100-95	94-85	84-75	74-65	64-60	<60
	对应绩点	5	课程成绩÷10-4.5				0

五级制	成绩	优秀	良好	中等	及格	不及格
	对应绩点	4.5	3.5	2.5	1.5	0

（二）课程学分绩点=课程绩点×课程学分数

（三）平均学分绩点（GPA）=符合条件的课程学分绩点之和÷相同条件的课程学分之和

平均学分绩点计算的课程范围为全部课程，计算结果用于评价同一段内学生学习质量的优劣。





青島理工大學
QINGDAO UNIVERSITY OF TECHNOLOGY

Proof of Grade Point Average and Ranking

This is to certify that Zhang Cenxi, male, Student ID: 202301050482, is an undergraduate student in the Class of 2023 majoring in Building Electrical and Intelligent at the School of Information and Control Engineering, Qingdao University of Technology. His current GPA for the courses he has completed is 4.55, and his grade ranking is 3/70. This certificate is only valid for the student's application for studying abroad.

Academic Affairs Office of Qingdao University of Technology
2026-05-15



Note:

1. Method for Converting Course Grade Points

Percentage System	Grade	100-95	94-85	74-65	64-60	<60
	Grade Point	5.0	Course grade $\div 10 - 4.5$			0

Five-tier System	Grade	Excellent	Good	Average	Pass	Fail
	Grade Point	4.5	3.5	2.5	1.5	0

2. Course Grade Point = Grade Point $\times$ Course Credits

3. Grade Point Average (GPA) = $\sum(\text{Course Grade Point}) \div \sum(\text{Course Credits})$

All courses are included in the GPA calculation, and calculation result is used to evaluate the quality of student learning in the same period.



青島理工大學
QINGDAO UNIVERSITY OF TECHNOLOGY

证明

兹证明我校课程绩点未设置 A、B、C、D 的等级。



Certificate

This is to certify that our university does not set levels of A, B, C, or D for Course Grade Points.

Academic Affairs Office of Qingdao University of Technology



Harnessed Agentic Simulation for Controllable Decision Support under Uncertainty

Cenxi Zhang

School of Information and Control Engineering
Qingdao University of Technology
Qingdao, Shandong, China

Abstract

This proposal studies a narrow problem: how an LLM-based agentic workflow can support financial portfolio scenario analysis without hiding assumptions, evidence, or intermediate reasoning. It does not aim to build a general autonomous agent or a financial prediction model. The proposed system, *Harnessed Agentic Simulation*, turns a portfolio decision question into a validated world specification, explicit evidence and assumptions, bounded simulation runs, provenance traces, and evidence-linked report claims. The central hypothesis is that such a workflow can improve traceability, verifiability, and auditability compared with one-shot LLM responses and simple prompt-only baselines. More broadly, the project asks how a long-horizon agentic workflow can remain stable when the model has only a limited context window.

1 Introduction

Financial decisions are often made with incomplete information. In portfolio analysis, a recommendation is only useful if the user can see which market assumptions were used, how risk was measured, and why one allocation is more robust than another. This makes traceability central rather than optional: investors and analysts need to inspect the evidence path behind a conclusion, check whether a result depends on a fragile assumption, and compare plausible futures before acting. Portfolio scenario analysis is therefore a suitable setting for decision support systems [8], decision analysis [6], and simulation-based reasoning.

Large language models can parse open-ended questions, summarize documents, and coordinate tools. However, a one-shot LLM response is weak for portfolio decisions because it usually merges evidence, assumptions, calculations, and recommendations into one fluent narrative. When the answer is wrong or unsupported, it is difficult to locate the failure point. This risk is related to hallucination and unsupported generation [7]. Existing agent work has advanced reasoning, acting, memory, coordination, and social simulation [11, 17, 18, 25, 29, 31], but it still leaves a practical gap for audited financial scenario analysis. This gap is especially important for long-running agents, where earlier assumptions and intermediate states may fall outside the active context window.

This proposal treats portfolio decision support as a staged workflow rather than a single answer-generation task. The system structures the decision, retrieves evidence, records assumptions, validates a WorldSpec, runs bounded scenarios, and generates claims that link back to evidence, assumptions, and trace events. The immediate research object is therefore precise: LLM-based agentic workflow design for traceable, verifiable, and auditable portfolio scenario analysis under uncertainty.

2 Research Gap and Questions

Existing LLM agent systems are usually optimized for task completion, dialogue, tool use, or benchmark performance. Classical agent-based modeling can represent interacting actors and emergent effects [3, 4, 15, 24], but traditional ABM often assumes that experts have already specified the model. Retrieval-augmented generation can ground answers in documents [10], and provenance standards can record data lineage [16]. Spec-driven AI development frameworks such as OpenSpec show a related move toward explicit proposals, requirements, designs, and task artifacts for coding agents [5], but they target software development workflows rather than executable decision-simulation worlds.

The specific gap is narrower: there is no clear workflow that converts a portfolio decision question into a validated WorldSpec and then connects evidence, assumptions, agent actions, state changes, metrics, and final report claims into one auditable chain. Current approaches can generate plausible scenarios, retrieve supporting documents, or log execution traces, but these pieces are rarely combined into a runnable and inspectable financial decision workflow that remains stable across many steps.

The key research problem is therefore not simply whether an LLM can imagine a scenario. It is whether a system can transform an ill-structured decision into an explicit, validated, executable, and inspectable artifact through staged orchestration. Compared with one-shot agent responses, this approach keeps evidence, assumptions, simulation state, agent actions, and final claims more clearly separated.

The proposal asks four research questions:

1. To what extent can a schema-based WorldSpec reduce invalid or incomplete portfolio scenarios compared with one-shot LLM scenario generation?
2. How effectively can evidence cards, assumption records, and provenance traces link final report claims to their supporting sources and simulation events?
3. Does a staged agentic workflow improve the clarity, inspectability, and decision usefulness of portfolio scenario reports compared with prompt-only and single-agent baselines?
4. What reliability costs are introduced by validation, repair, retrieval, and tracing, and which components contribute most to long-horizon workflow stability?

3 Proposed Method

The method is a staged workflow with explicit inputs, processing steps, and outputs. It is designed for portfolio scenario analysis, not general financial forecasting.

Problem structuring. The input is a natural-language portfolio question, such as whether to shift a moderate-risk portfolio toward equities under uncertain interest-rate and inflation conditions. The process extracts goals, constraints, candidate allocations, time horizon, risk tolerance, and evaluation metrics. The output is a DecisionSpec containing the decision objective, feasible interventions, and metrics such as expected return, drawdown, volatility, liquidity pressure, and capital exposure.

Evidence and assumption registration. The inputs are historical price data, market notes, and user-provided constraints. The process converts source-backed facts into EvidenceCards and unsupported modeling choices into editable Assumptions. The output is a small evidence and assumption registry that separates observed information from choices made for the simulation.

World-generation harness. The inputs are the DecisionSpec, EvidenceCards, and Assumptions. The process generates a constrained WorldSpec with asset groups, stakeholder agents, scenario parameters, interventions, update rules, metrics, and source links. Before execution, the harness checks schema validity, missing references, parameter ranges, unsupported claims, and executable constraints. This follows the simulation principle that verification and validation should be built into modeling [23]. The output is either a valid runnable WorldSpec or a repair request with explicit error messages.

Controlled orchestration. The inputs are validated or repairable artifacts from earlier stages. The process routes work through small modules: scenario drafting, validation, repair, compilation, execution, retrieval, tracing, and report generation. This design follows work on transparent chained

workflows [30] and graph-structured agentic workflow evaluation [20]. The output is an execution record in which each module has a defined responsibility and observable failure state.

Runtime, tracing, and reporting. The inputs are the compiled WorldSpec and selected scenarios. The runtime executes scenario steps with stakeholder agents representing an investor, analyst, and risk manager. At each step, agents observe current metrics, retrieve relevant evidence, take bounded actions, and update shared state. The output is a visual timeline, a machine-readable provenance trace, and a report whose claims must link to evidence, assumptions, and trace events, extending standard provenance toward agentic workflows [26].

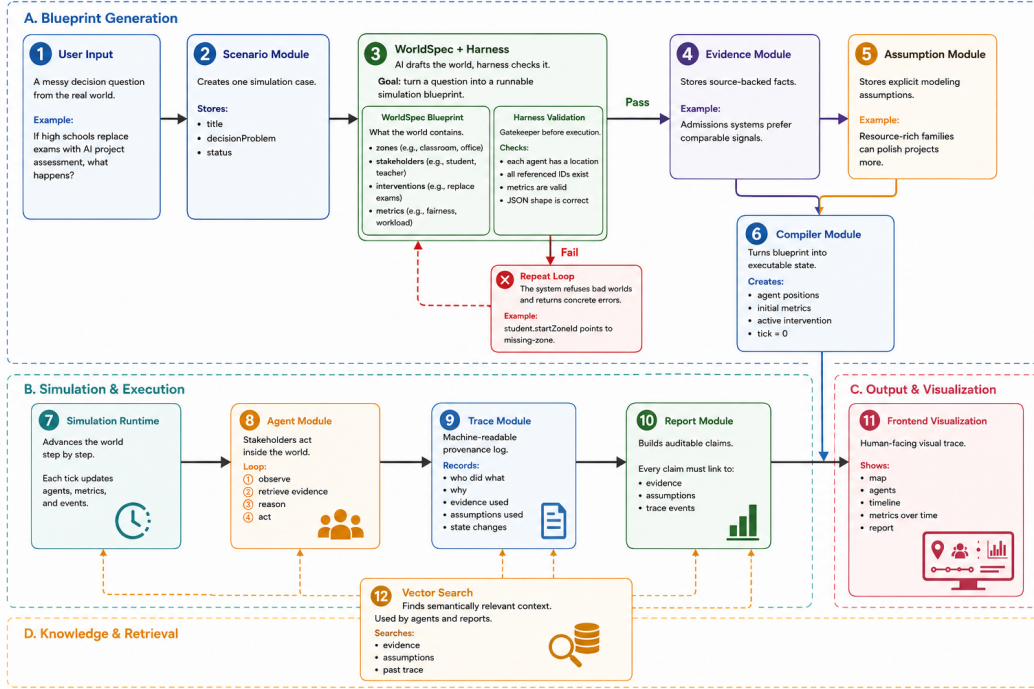


Figure 1: Harnessed agentic simulation architecture for financial scenario generation and traceable decision support. The workflow decomposes a financial decision question into scenario creation, world-specification drafting, harness validation, repair, evidence and assumption registration, compilation, stepwise simulation, agent action, provenance tracing, evidence-linked reporting, frontend visualization, and vector-search retrieval.

Figure 1 shows the concrete orchestration graph. Its main role is to force intermediate artifacts into the open: bad worlds are rejected before execution, missing evidence remains visible, assumptions are editable, and final claims require trace support. The core objects are explicit data structures rather than prompt text, making the system inspectable and testable.

Object	Required Fields	Purpose
WorldSpec	zones, agents, interventions, metrics, rules	Runnable simulation blueprint.
EvidenceCard Assumption	source, claim, scope, reliability statement, parameter, uncertainty, editable flag	Source-backed factual grounding. Explicit modeling choice.
TraceEvent	tick, actor, action, state change, links	Replayable provenance record.
ReportClaim	claim, evidence links, assumption links, trace links	Auditable conclusion.

Table 1: Core structured objects used by the harnessed workflow.

4 Main Example and Evaluation

The main case is a moderate-risk portfolio reallocation problem over a 12-month horizon. The prototype will use broad asset classes rather than individual stock picking: U.S. equities, aggregate bonds, gold, and cash or money-market exposure. The market scenarios will include a base case, an inflation-and-rate-pressure case, and an equity drawdown case. This level of detail keeps the case financially recognizable while avoiding unsupported claims about precise market prediction.

The system will compare three candidate allocations: a growth allocation with higher equity exposure, a balanced allocation, and a conservative allocation with more bonds, gold, and cash. Stakeholder agents represent an investor, an analyst, and a risk manager. Metrics include scenario return, maximum drawdown, volatility, liquidity pressure, and exposure concentration. The case is grounded in decision analysis, robust scenarios, and simulation-based reasoning [6, 9, 27]. An open-source demo already demonstrates the core pipeline and visualization layer [34].

Evaluation will compare four systems: a direct prompt-only LLM answer, a single tool-using LLM agent, the full harnessed workflow, and ablated workflow variants without validation, without retrieval, or without provenance tracing. The main quantitative measures are WorldSpec validity rate, consistency of intermediate state across steps, number of unsupported claims, percentage of report claims linked to evidence and trace events, repair-loop success rate, and runtime overhead. A small human evaluation will compare reports on clarity, inspectability, and usefulness for decision support, following recent work on realistic and fine-grained agent evaluation [13, 14, 21, 32, 33, 35].

5 Research Timeline

The project is scoped as an 18-month research plan. The goal is not to claim a general theory of all autonomous agents, but to deliver one well-specified orchestration framework, one runnable prototype, one detailed finance case study, and one focused evaluation.

Phase	Focus	Output
Months 1–3	Literature review; research questions; system scope.	Schema and requirements.
Months 4–6	Scenario, evidence, assumption, and WorldSpec modules.	Validated WorldSpec and repair-loop prototype.
Months 7–9	Compiler, runtime, agent loop, trace logging, and vector search.	Runnable finance scenario simulation.
Months 10–12	Frontend visualization and auditable report generation.	Visual timeline, metrics, traces, and linked report.
Months 13–15	Evaluation against prompt-only and single-agent baselines.	Traceability, validity, ablation, and error analysis.
Months 16–18	Framework refinement, thesis writing, and paper preparation.	Final thesis, reproducible demo, and paper draft.

Table 2: Planned research timeline and deliverables.

6 Expected Outcomes and Value

The expected outcome is not only a prototype, but reusable infrastructure for studying complex decisions with long-running agents. The project addresses a missing layer between flexible LLM agents and rigorous decision-support simulation: a structured workflow in which evidence, assumptions, intermediate states, and final claims remain inspectable.

The value is threefold. For AI-agent research, the project studies how validation, repair loops, provenance, retrieval, and modular orchestration can make long-running agents more reliable than one-shot answer generation. For decision-support research, it offers a way to compare scenarios under uncertainty without hiding the assumptions behind a recommendation. For interdisciplinary research, it provides a reusable workflow for domains where real-world trial and error is costly, slow, or ethically difficult.

Without such infrastructure, agentic decision support may remain fluent but hard to trust: users may receive plausible recommendations without knowing which evidence was used, which assumptions shaped the result, whether the simulated world was valid, or how errors propagated through the workflow.

The concrete outcomes are:

1. A compact orchestration schema that makes goals, evidence, assumptions, interventions, world specifications, intermediate states, and outcomes explicit.
2. A small but complete harnessed agentic simulation pipeline for scenario comparison under uncertainty.
3. A validation and repair framework for preventing AI-generated world specifications from entering execution when they are incomplete or invalid.
4. A dual trace layer that records human-facing visual traces and machine-readable provenance traces for auditability.
5. A focused case study in finance showing how traceable decision simulation can support practical discussion without relying on a single one-shot agent answer.
6. An open-source demo implementation that makes the proposed workflow reproducible and inspectable.

7 Related Work

LLM-agent research has advanced reasoning, acting, reflection, multi-agent collaboration, and agent evaluation [11, 13, 25, 29, 31]. Recent surveys further organize this area around autonomous-agent architectures, agentic LLM capabilities, and multi-agent workflow infrastructure [12, 19, 28]. Generative Agents and AI Town show how LLM-driven agents can inhabit persistent worlds [1, 18]. This proposal builds on these systems but shifts the goal from open-ended behavior to controlled, auditable scenario comparison.

The project also draws on agent-based modeling, robust decision-making, retrieval, and provenance. ABM explains how local interactions generate system-level effects [2–4, 15, 24], while robust scenario methods address decision-making under deep uncertainty [9, 27]. Retrieval-augmented generation grounds outputs in documents [10]; W3C PROV and PROV-AGENT support traceable links among activities, agents, prompts, decisions, and downstream effects [16, 26]. The gap is an integrated framework that connects these ideas into a validated agentic workflow for finance with evidence-linked reports.

8 Challenges

Three challenge-strategy pairs shape the project. First, generated simulations can look plausible without being decision-useful. The strategy is to avoid prediction claims and instead evaluate traceability, assumption visibility, and sensitivity under alternative scenarios [9, 22, 27]. Second, agentic workflows add orchestration cost. The strategy is to measure runtime overhead and use ablations to test whether validation, repair, retrieval, and tracing justify their cost. Third, the

framework may not generalize beyond the first case. The strategy is to begin with one concrete finance case, define the reusable schema clearly, and treat any later cross-domain use as an extension rather than an initial claim.

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